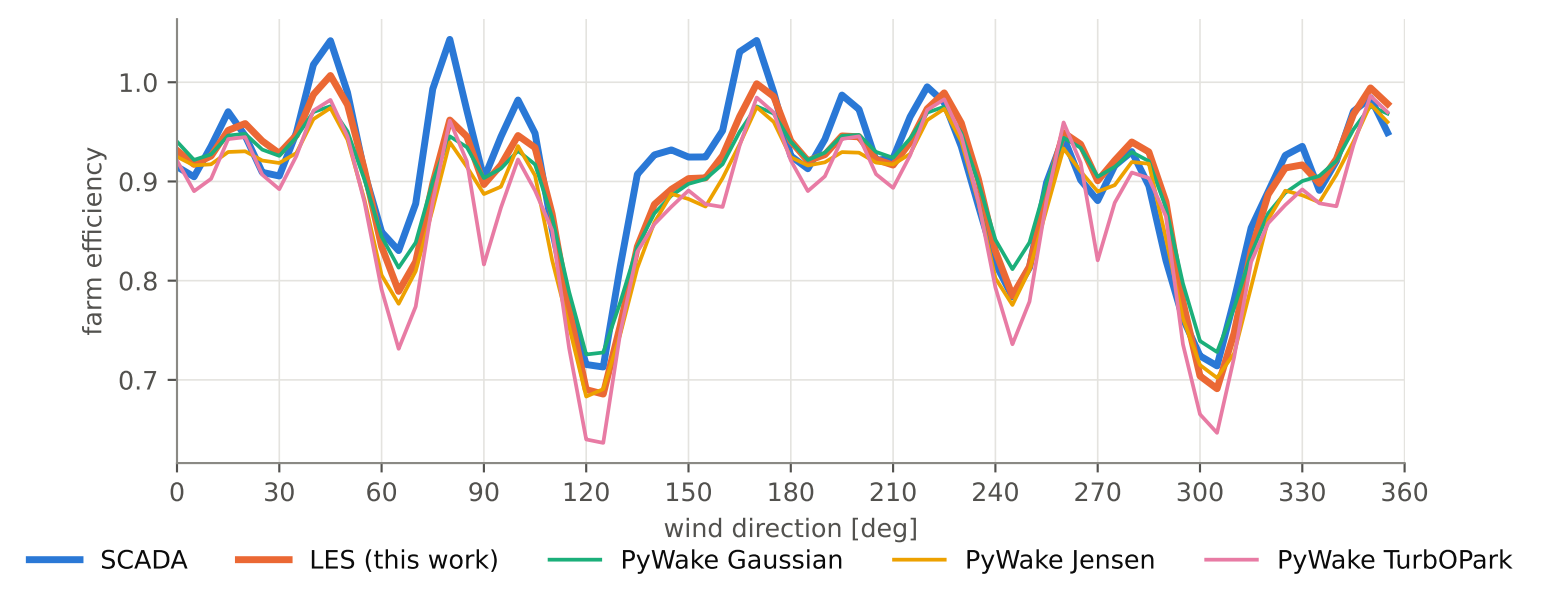
farm efficiency
1.0
0.9
0.8
0.7
0
30
60
90
120
150
180
210
240
270
300
330
360
wind direction [deg]
SCADA
LES (this work)
PyWake Gaussian
PyWake Jensen
PyWake TurbOPark